



Total RNA-Seq Data Analysis Report

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Contents

1	RNA Library Preparation.	3
2	Executive Data Analysis Summary.	3
3	Methods – Data Analysis	4
3.1	Bioinformatics Pipeline	4
3.2	Read quality check	5
3.3	Read alignment	5
3.4	Alignment quality screening.	5
3.5	Expression estimation	6
3.6	Differential expression analysis	6
3.7	Functional Gene Set Enrichment Analysis.	6
3.8	Samples and Grouping information	6
4	Results.	7
4.1	Data QC Summary	7
4.2	Alignment Summary	7
5	Sample dimensional reduction and Clustering	8
5.1	Principal Component Analysis (PCA)	8
5.2	Hierarchical Clustering	8
5.3	Sample Correlation	9
6	Differential Gene Expression (DGE) Analysis.	10
7	Representation of DEGs as plots.	10
7.1	Volcano plots.	10
8	Functional Gene Set Enrichment Analysis	11
9	Coexpression Analysis of known lncRNA-mRNA.	12
10	lncRNA-mRNA Target Analysis	13
11	lncRNA Functional Enrichment Analysis	13

12	LncRNA-mRNA Network	13
13	Novel lncRNA Analysis	13
14	Deliverables	14

1 RNA Library Preparation

Paxgene miRNA extraction protocol centrifuging the PAXgene Blood RNA Tube at $\sim 5,000 \times g$ for 10 minutes, discarding the supernatant, and resuspending the pellet in 4 ml RNase-free water followed by a second centrifugation and removal of the supernatant. The pellet is then resuspended in 350 μ l Buffer BR1, transferred to a microcentrifuge tube, and mixed with 300 μ l Buffer BR2 and 40 μ l proteinase K, followed by incubation at 55 °C for 10 minutes. After lysis, 350 μ l of 100% ethanol is added and mixed thoroughly, and the lysate is loaded onto a spin column and centrifuged. The column is washed with 500 μ l Buffer BR3, followed by on-column DNase digestion using 10 μ l DNase I + 70 μ l Buffer RDD (total 80 μ l), incubated for 15 minutes at room temperature. Subsequent washes include 500 μ l Buffer BR3 and 500 μ l Buffer BR4 with centrifugation steps. Finally, the column is dried by an additional spin and RNA is eluted in 30 μ l RNase-free water. Samples are quantified using the Agilent TapeStation 4150 RNA Screen Tapes (5067-5576) and Invitrogen Qubit 4 HS RNA assay(Q32855).

rRNA depletion using the QIAseq FastSelect -rRNA/Globin Kit (335377). To perform depletion, 500ng of total RNA is mixed with FastSelect reagent and denaturation buffer, then incubated through a short thermal protocol (e.g., 70°C for 5 minutes, then ramping down to 25°C) to allow hybridization of FastSelect oligos to rRNA targets. After rRNA depletion, total RNA prepared using NEBNext® Ultra™ II Directional RNA Library Prep Kit for Illumina(E7760L). RNA fragmentation and priming are carried out by incubating the RNA with NEBNext First Strand Synthesis Buffer and Random Primers at 94°C for 8 minutes. First strand cDNA synthesis follows using reverse transcriptase at 25°C for 10 minutes, 42°C for 15 minutes, and 70°C for 15 minutes. Second strand synthesis incorporates dUTP for strand specificity and proceeds at 16°C for 1 hour, followed by cleanup with AMPure XP beads. End repair and dA-tailing are performed by incubating with End Prep reagents at 20°C for 30 minutes and 65°C for 30 minutes. Adapter ligation is carried out at 20°C for 15 minutes, followed by USER enzyme treatment at 37°C for 15 minutes. After the bead purification, PCR enrichment (8 cycles) is performed with NEBNext Q5 Master Mix and indexed primers. The final library is cleaned, assessed for fragment size using the Agilent TapeStation 4150 and Invitrogen Qubit 4 HS DNA assay(Q32851)Quantified libraries sequenced using Illumina Novaseq 6000.

2 Executive Data Analysis Summary

We analyzed a total of 9 Human samples, as shared by the client. Samples and Grouping information are provided in Table 1 and Table 2. The raw reads were filtered using Cutadapt for quality scores and adapters. Filtered reads were aligned to the Human genome (GRCh38.primary_assembly.genome) using splice aware aligner STAR to quantify reads mapped to each gene. Percentage of uniquely aligned reads ranged between 87% – 91% across all the samples. The aligned data quality check was performed using Qualimap. rRNA contamination was screened using RSeqQC package. Total number of uniquely mapped

reads were counted using FeatureCounts. The uniquely mapped reads were then subjected to differential gene expression using DESeq2 in R. Total number of differentially expressed genes are given in the Table 5 for the respective comparisons.

3 Methods – Data Analysis

3.1 Bioinformatics Pipeline

The following bioinformatics steps were performed for analysis of the data.

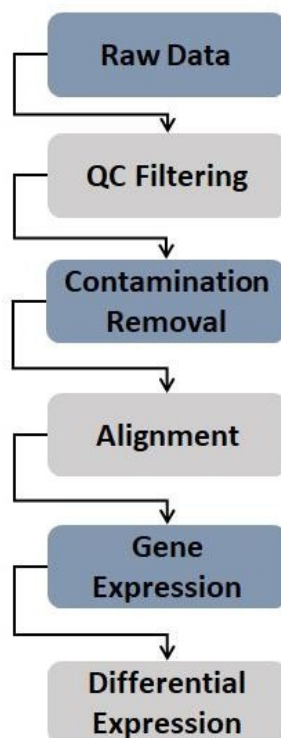


Figure 1: Bioinformatics workflow for RNA-seq analysis

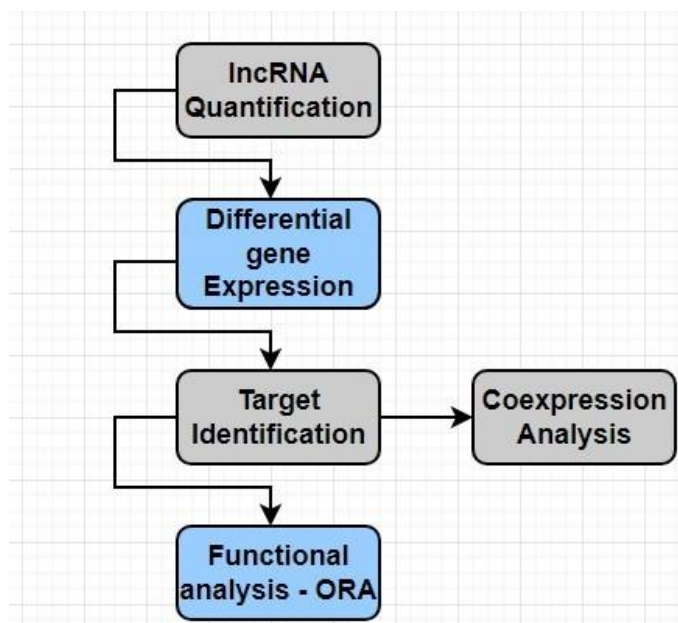


Figure 2: Bioinformatics workflow for known lncRNA analysis

3.2 Read quality check

The following parameters from raw fastq files were checked using fastqc tool (version – 0.11.9) as part of quality check- Base quality score distribution, Sequence quality score distribution, Average base content per read, GC distribution in the reads, PCR amplification issue, Check for over represented sequences and Adapter trimming. Based on the quality report of fastq files trimming on raw read was performed to only retain high quality sequence for further analysis. In addition, the low-quality sequence reads were excluded from the analysis. The adapter trimming was performed using Cutadapt (Version 4.9).

3.3 Read alignment

The paired-end reads are aligned to the reference Human genome (GRCh38.primary_assembly.genome). Alignment of reads was performed using STAR (version - 2.7.11b)

3.4 Alignment quality screening

Aligned reads were analyzed for their quality check related statistics; against both mapping quality as well as read alignment distribution against reference transcriptome features using Qualimap. Factors like read distribution, Alignment distribution as well as splice junction distribution were analyzed separately. rRNA contamination was screened using rSeqQC package.

3.5 Expression estimation

The aligned reads were used for estimating expression of the genes. The raw read counts were estimated using FeatureCount (version - 2.0.8) across all samples. The raw read count across all samples were then normalized using DESeq2 package in R.

3.6 Differential expression analysis

The raw read count data were normalized using DESeq2. The ratio of normalized read counts for test over control was taken as the fold change. Genes were first filtered based on the p-value (≤ 0.05) for statistical significance. Those genes which were found to have $\log_2(\text{foldchange}) \leq -1$ and $\log_2(\text{foldchange}) \geq 1$ were considered as downregulated or upregulated, respectively.

3.7 Functional Gene Set Enrichment Analysis

Functional gene set enrichment analysis was performed for each comparison using different approaches for lncRNA and mRNA datasets. For lncRNA-associated genes, Over-Representation Analysis (ORA) was carried out using the clusterProfiler R package to identify significantly enriched functional categories among differentially expressed genes. For mRNA expression data, Gene Set Enrichment Analysis (GSEA) was performed using ranked gene expression profiles to detect coordinated enrichment patterns without applying a strict differential expression cutoff. Enrichment analyses were conducted against multiple ontology and pathway databases, including Gene Ontology categories such as Biological Process (BP), Molecular Function (MF), and Cellular Component (CC), along with Reactome pathways and Hallmark gene sets. These analyses provide insights into the activated or suppressed biological functions and pathways in the Test group relative to the Control group.

3.8 Samples and Grouping information

Following samples were analyzed.

Table 1: Samples Information

theraCUES Sample ID	Sample Name
SAM8451	NC-002
SAM8452	NC-003
SAM8453	NC-004
SAM8454	NC-005
SAM8455	GDM-001
SAM8456	GDM-002
SAM8457	GDM-003
SAM8458	GDM-004
SAM8459	GDM-005

4 Results

4.1 Data QC Summary

Below table summarizes the overall data generated and the average read quality observed across the reads.

Table 2: Raw Data QC Summary

theraCUES Sample ID	Sample Name	Data generated (Gb) - before trimming	Avg. base quality (phred) before trimming	% of reads belonging to Human	Total paired reads (After adapter trimming)	Percentage of reads remaining after trimming (%)	Avg. base quality (phred) after trimming	GC% after trimming
SAM8451	NC-002	12	39	99	39568460	99	40	43
SAM8452	NC-003	11	39	98	35816826	99	40	44
SAM8453	NC-004	10	39	94	32141911	99	40	43
SAM8454	NC-005	9	39	98	31095498	99	40	44
SAM8455	GDM-001	9	39	97	29644435	99	40	47
SAM8456	GDM-002	10	39	98	32797386	99	40	46
SAM8457	GDM-003	10	39	98	33120605	99	40	45
SAM8458	GDM-004	12	39	98	39092045	99	40	44
SAM8459	GDM-005	11	39	99	36872502	99	40	44

Data generation for each sample was in the range of 9 GB to 12 GB and the average % of reads with quality > phred score 30 was 99 %.

4.2 Alignment Summary

Samples were aligned to 91-95% reference genome. The mapping statistics showed the alignment of uniquely mapped reads were at the range of 87 to 92%. The below Table summarizes the alignment summary of the sequenced reads.

Table 3: Mapping Statistics Summary

theraCUES Sample ID	Sample Name	Uniquely mapped reads %	Reads Unaligned (%)	Total mapped read%	Exonic (%)	Intronic (%)	Intergenic (%)	Rseqc rrna estimation (%)
SAM8458	GDM-004	88	7	93	21	65	14	1
SAM8453	NC-004	88	8	92	32	65	3	1
SAM8456	GDM-002	89	5	95	34	62	3	2
SAM8457	GDM-003	90	6	95	33	64	3	1
SAM8455	GDM-001	90	5	95	37	60	3	3
SAM8452	NC-003	91	5	95	30	65	4	0
SAM8459	GDM-005	92	4	96	32	65	4	1
SAM8454	NC-005	92	4	96	37	59	4	1
SAM8451	NC-002	92	5	96	26	66	8	0

5 Sample dimensional reduction and Clustering

5.1 Principal Component Analysis (PCA)

Principal Component Analysis (PCA) is a statistical technique used to identify global patterns in high-dimensional datasets. It is commonly used to explore the similarity of biological samples in RNA-seq datasets. To achieve this, gene expression values are transformed into Principal Components (PCs), a set of linearly uncorrelated features which represent the most relevant sources of variance in the data, and subsequently visualized using a scatter plot.

The scatter plot of the first two principal components (pcs) of the data for given samples are shown in below Figure. Each point represents the sample. Samples with similar gene expression profiles should come closer in the two-dimensional space.

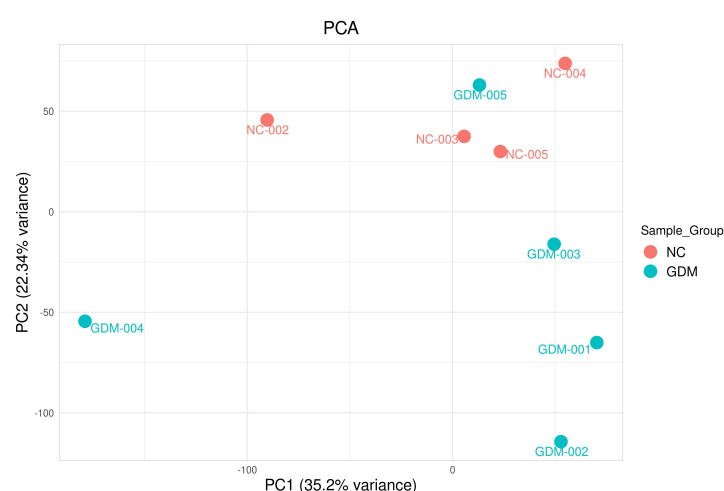


Figure 3: PCA plot: Represents the scatter plot of the first two principal components of the samples

5.2 Hierarchical Clustering

Clustering is a statistical technique used to identify the similarity between data. We used hclust R function together with average linkage and euclidean distance to perform clustering, samples whose gene expression are similar try to cluster together as shown in below Figure. Ideally, samples should cluster according to the group i.e Samples belonging to the same group must be clustered together.

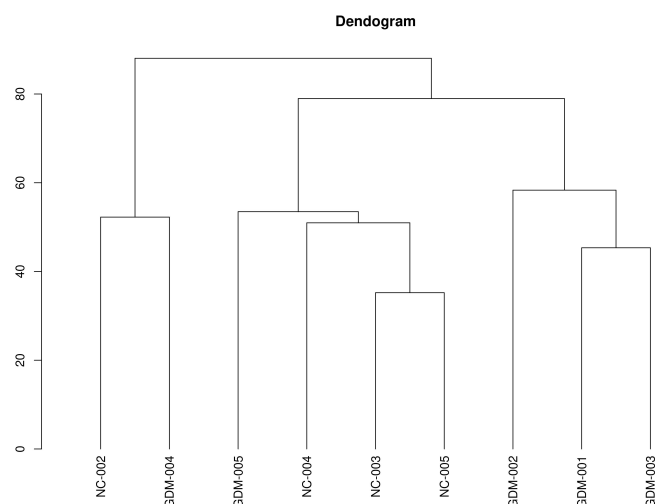


Figure 4: Represents the hierarchical clustering of the data for given samples

5.3 Sample Correlation

A sample-sample correlation heatmap shows how similar samples are based on their gene expression. High correlation among replicates indicates similarity, while low correlation can reveal outliers and clustering of samples based on their type determines their similarity.

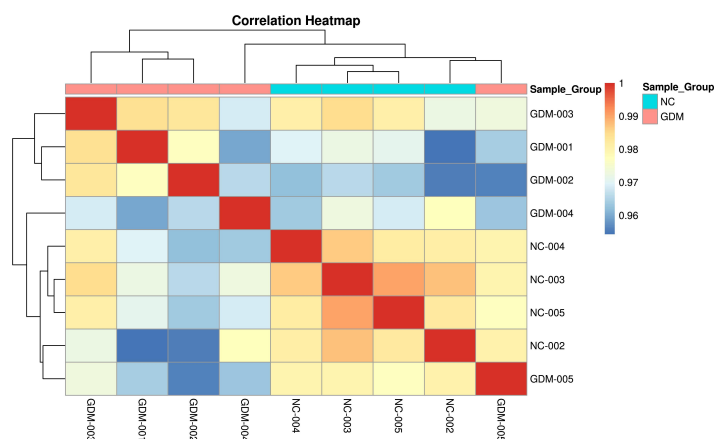


Figure 5: Represents correlation of samples along with its replicates

6 Differential Gene Expression (DGE) Analysis

Gene expression signatures are alterations in the patterns of gene expression that occur due to cellular perturbations such as drug treatments, gene knockdown or diseases. They can be quantified using differential gene expression (DGE) methods, which compare gene expression between two groups of samples to identify genes whose expression is significantly altered in the perturbation. The signature table is used to display the results of such analyses.

Table 4: Comparisons performed in Differential Gene Expression (mRNA) analysis

Comparison	Group
1	GDM vs NC

Table 5: Summary of Differentially Expressed Genes [mRNA](DEGs)

Comparison	No_of_upregulated_genes	No_of_downregulated_genes
GDM vs NC	578	47

Table 6: Comparisons performed in Differential Gene Expression (lncRNA) analysis

Comparison	Group
1	GDM vs NC

Table 7: Summary of Differentially Expressed lncRNA

Comparison	No_of_upregulated_genes	No_of_downregulated_genes
GDM vs NC	349	92

7 Representation of DEGs as plots

7.1 Volcano plots

Volcano plots are a type of scatter plot commonly used to display the results of a differential gene expression analysis. They can be used to quickly identify genes whose expression is significantly altered in a perturbation, and to assess the global similarity of gene expression in two groups of biological samples. Each point in the scatter plot represents a gene; the axes display the significance versus fold-change estimated by the differential expression analysis. Green points indicate significantly downregulated and red points indicate significantly upregulated genes and grey ones represent statistically non-significant genes.

Every dot in the plot represents a gene. Green points indicate significantly downregulated genes and red points indicate significantly upregulated genes.

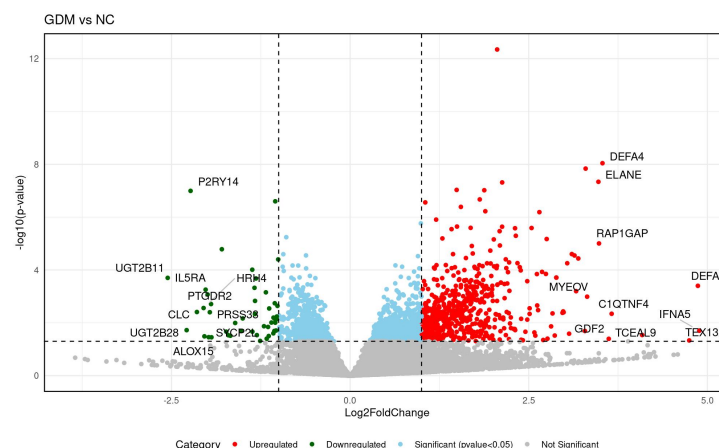


Figure 6: Volcano plot for the given GDM vs NC comparison showing log2 fold change and statistical significance of each gene

8 Functional Gene Set Enrichment Analysis

Functional regulation associated with deregulated genes across different sample groups was analysed using pathway and functional enrichment approaches tailored for mRNA and lncRNA-associated targets. For mRNA expression profiles, Gene Set Enrichment Analysis (GSEA) was performed using ranked gene expression data to identify coordinated enrichment patterns without relying solely on differential expression cutoffs. Enrichment analysis was carried out against public functional databases and ontologies including Gene Ontology categories Biological Process (BP), Cellular Component (CC), and Molecular Function (MF) along with Reactome and Hallmark pathway databases. The top significantly enriched pathways identified through GSEA are represented as enrichment plots and summarized for biological interpretation.

Pathways that are significantly enriched in a gene list are shown as Barplot. Analysis of these functional categories revealed the association of the differentially expressed genes in activation and suppression of multiple functions/pathways.

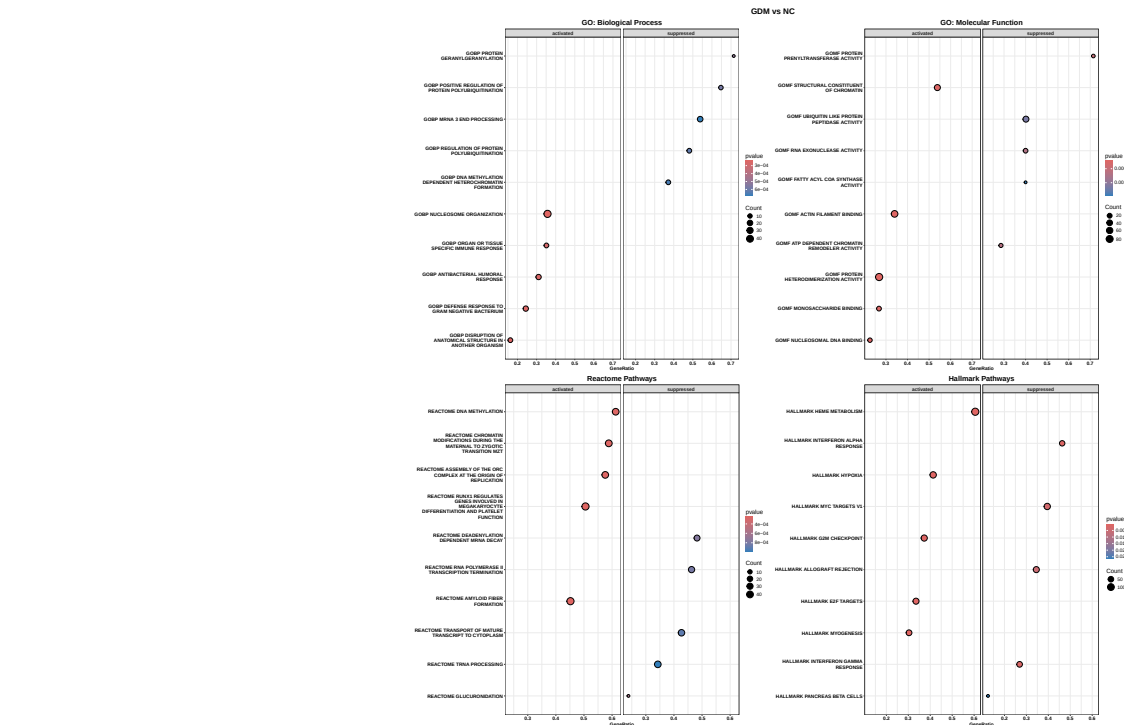


Figure 7: Functional over representation analysis of GDM vs NC comparison showing upregulated mRNA pathways (top 5)

All visualizations and analytical plots generated for the mRNA datasets were similarly produced for the known lncRNA datasets and organized within their respective corresponding folders.

9 Coexpression Analysis of known lncRNA-mRNA

The lncRNA and mRNA expression counts obtained from the same set of samples were further analysed using Pearson correlation analysis to identify potential lncRNA–mRNA co-expression relationships and predict regulatory targets. Correlation pairs showing statistical significance ($p < 0.05$), irrespective of correlation strength, were considered as significant co-expression pairs for downstream analysis.

10 LncRNA-mRNA Target Analysis

Significantly correlated lncRNA-mRNA pairs were further analysed to identify potential cis-regulatory targets. Classification of lncRNA-associated mRNA targets into cis and trans categories based on genomic co-localization was performed using the standalone version of FEELnc software.

11 LncRNA Functional Enrichment Analysis

For significantly co-expressed lncRNA-mRNA pairs, functional enrichment analysis was performed on the associated mRNA target gene sets using the Over-Representation Analysis (ORA) method to identify biological functions and pathways significantly enriched among the correlated targets. ORA was conducted against Gene Ontology categories, Reactome pathways, and Hallmark pathway databases.

12 LncRNA-mRNA Network

Based on the number of correlated mRNA targets, the top 10 hub lncRNAs were identified. For network construction, the top 10 lncRNAs along with their top 20 correlated mRNA targets, selected based on absolute fold change values, were used to generate the lncRNA-mRNA co-expression network. This analysis highlights potential regulatory interactions and provides insights into key lncRNAs associated with transcriptional regulation across the studied conditions.

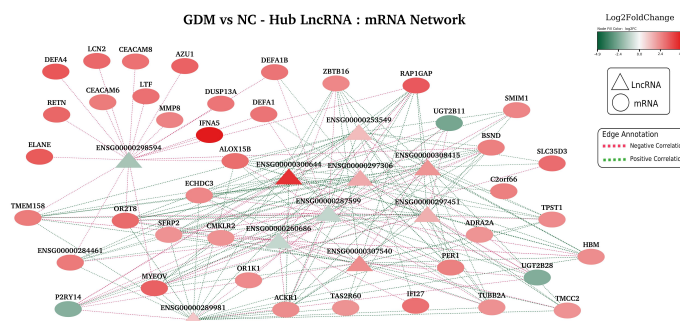


Figure 8: Top 10 LncRNA - mRNA Interaction Network:

13 Novel lncRNA Analysis

The mapped reads of each sample were re-assembled with StringTie and merged with merge command of StringTie. Further to identify known lncRNA a combined database of lncRNAs from NONCODE, lncRNAKB and Gencode was created and searched against the re-assembled

transcripts using CuffCompare, to obtain novel lncRNA transcripts that did not match known annotated genes. Three types of transcripts (I, u, and x) were further extracted for lncRNA prediction, where i indicates transfrags falling entirely within a reference intron, u indicates unknown and intergenic transcripts, and x refers to exonic overlap with reference on the opposite strand. Then, transcripts with a length greater than or equal to 200 bp, more than two exons, and ORF less than 300 were chosen. Further CPAT and PLEK were utilised for evaluating the coding potential of the transcripts and the intersection of their predicted results was obtained. After excluding known lncRNA sequences, the remaining transcripts were searched against the Pfam database. Transcripts whose translated frames showed no significant BLAST hits, along with those without any BLAST matches, were considered as novel lncRNAs. Subsequently, the expression counts of these lncRNAs were quantified using featureCounts.

Classification of novel lncRNA-associated mRNA targets into cis and trans categories based on genomic co-localization was performed using the standalone version of FEELnc software.

All the corresponding plots related to all the comparisons and other analyses are uploaded in the corresponding folders. Please refer to the respective comparison folders for volcano plots, functional enrichment bar plots, PCA plots, hierarchical clustering dendrograms, and sample-to-sample correlation heatmaps.

14 Deliverables

Standard deliverables:

1. Raw Data QC and Raw Data counts across all samples.
2. Normalized read count for each sample along with Sample Grouping/Clustering and QC related plots: PCA plot, Heatmap based on top 2000 variant genes, dendrogram and Sample-Sample correlation plot.
3. Differential expression reports for each comparison along with corresponding visualization – Volcano plots.
4. Functional over-representation (enrichment) analysis reports per comparison, along with corresponding visualization – Gradient Bar plots for top 20 enriched GO: BP, MF, CC and Reactome Pathways.
5. Raw Count and Normalized count of known lncRNA for each sample along with Sample Grouping/Clustering and QC related plots: PCA plot, Heatmap based on top 2000 variant genes, dendrogram and Sample-Sample correlation plot.
6. Differential expression reports of known lncRNA for each comparison along with corresponding visualization – Volcano plots.
7. Functional over-representation (enrichment) analysis reports per comparison, along with corresponding visualization – Gradient Bar plots for top 20 enriched Reactome Pathways for Positively & Negatively Correlated Upregulated mRNAs and Positively & Negatively Correlated Downregulated mRNAs.

8. Target prediction and FEELnc employing DEIncRNA and DEmRNA , and DEIncRNA and DEmRNA for FEELnc; Interaction type (Cis or Trans) of each lncRNA.
9. lncRNA-mRNA network.
10. Novel lncRNAs rawcounts and Cis mRNA interaction

Access to all reports:

(<https://workdrive.zohoexternal.com/external/cf61a32f2f0df1b4c335714b5be5295935c79669af62b3c931ea13b2aec802f8>)